

Lab_task

Intake 53/1

1. You are designing a system where different types of vehicles have their own way of starting. Create a parent class `Vehicle` with a method `start()`. Create subclasses `Car` and `Bike` that override the `start()` method. Write a program to demonstrate dynamic method dispatch when calling `start()` using a `Vehicle` reference.
2. Suppose you are creating a simulation of animal sounds. Define a class `Animal` with a method `makeSound()`. Create subclasses `Dog` and `Cat` that override `makeSound()`. Demonstrate dynamic method dispatch by calling `makeSound()` using an `Animal` reference.
3. You are developing an e-commerce payment system. Create a class `Payment` with a method `pay()`. Create subclasses `CreditCardPayment` and `PayPalPayment` that override `pay()`. In the `main()` method, use a `Payment` reference to demonstrate runtime polymorphism.
4. Design a class hierarchy for shapes. Create an abstract class `Shape` with a method `draw()`. Create subclasses `Circle` and `Rectangle` that override `draw()`. Write a program to call `draw()` on a `Shape` reference pointing to different objects.
5. In a company, different employees have different salary structures. Create a class `Employee` with a method `calculateSalary()`. Subclass `FullTimeEmployee` and `PartTimeEmployee` should override `calculateSalary()`. Demonstrate dynamic method dispatch in the `main()` method using an `Employee` reference.
6. An abstract class called **Marks** is needed to calculate the percentage of marks earned by students A in three subjects (with each subject out of 100) and student B in four subjects (with each subject out of 100). This class must contain the abstract method “`getPercentage`,” which two other classes, “A” and “B,” will inherit. The method “`getPercentage`,” which provides the percentage of students, is shared by classes “A” and “B.”
The constructor of class ‘A’ will accept the marks obtained in three subjects as its parameters and the constructor of class ‘B’ will accept the marks obtained in four subjects as its parameters. To test the implementation, objects for both the classes need to be created and the percentage of marks for each student should be printed.